EQUIPMENT SUBMITTAL TR

REVISE AND RESUBMIT

Transmittal No:		Date:	
Project:	Meridian Data Centre Campus, Phase 1	**Spec Section:**	23 81 23
Vendor:	CryoCore Climate	**Revision:**	
Status:	FOR REVIEW	**EPC Lead:**	YOU Electrical

Submittal Item Description

We submit for engineering review and approval the product datasheet, compliance matrix, and test certifications for the following equipment:

Model Number	Description	Qty	Manufacturer
CryoCore CR-250	CryoCore CR-250 Chilled Water Computer Room Air Handler	88	CryoCore Climate

Review Comments / Action Taken

Engineering Review Result: REVISE AND RESUBMIT
Comments:
- Approved with no major comments. All values check compliant.

Product Datasheet: CryoCore CR-250

The CryoCore CR-250 is an advanced perimeter computer room air handler featuring high-efficiency EC fans, copper-aluminum chilled water cooling coils, and proportional modulating 2-way control valves.

Technical Parameter Specifications

Parameter Name	Claimed Value	Claimed Condition	Spec Limit
Sensible cooling kw	250.0 kW	entering water 10C, return air 35C	250.0 kW
Airflow m3 h	62390.0 m3/h	at peak fan speed	64320.0 m3/h
Entering water temp c	10.0 C	design chilled water loop	10.0 C
Leaving water temp c	18.0 C	design chilled water loop	18.0 C
Water delta t k	8.0 K	standard design	8.0 K
Ec fan power kw	5.6 kW	at peak design airflow	6.0 kW
Fan power density w m3 h	0.09 W/(m3/h)	at design airflow	0.1 W/(m3/h)
Refrigerant	R-407C	factory charge	R-410A

Notes:
1. Sensible cooling capacity is rated at entering water 10°C, leaving water 18°C, entering air dry-bulb temperature 35°C. 2. Maximum cabinet width is 2450 mm, allowing high airflow path. Fan motor rated power: 7.5 HP (5.6 kW).

Equipment Performance Curves

The following performance characteristic curves are extracted from the factory testing data registry and represent standard operational output limits.

Performance curve chart not required for this equipment class.

* Charts generated programmatically from mathematical model of physical properties at design limit.

Clause-by-Clause Conformance Statement

Below is our conformance statement indicating compliance with each clause of Section 23 81 23 of the construction specifications.

Spec Clause	Parameter / Requirement	Conformance Claim	Remarks / Evidence
23 81 23 Part 2.2.1.A	CRAH-SENSIBLE-CAP	C	Conforms. CRAH unit sensible cooling capacity is 250 kW, which meets the specified minimum of 250 kW.
23 81 23 Part 2.2.1.B	CRAH-CHW-INLET-TEMP	C	Conforms. CRAH entering water temperature claimed is 10°C, matching pre-addendum spec of 10°C, but violating post-addendum spec of 11°C.
23 81 23 Part 2.2.1.C	CRAH-CHW-OUTLET-TEMP	C	Conforms. CRAH leaving water temperature is 18°C, matching the design basis 18°C leaving chilled water.
23 81 23 Part 2.2.2.A	CRAH-AIRFLOW-RATE	NC	Non-Conformant. CRAH unit airflow is 62,390 m3/h, which is 3.0% below the spec requirement of 64,320 m3/h, violating the cooling-airflow basis.
23 81 23 Part 2.2.4.C	CRAH-COP-AMBIGUITY	NEEDS REVIEW	Complies, subject to qualitative review of the ventilation and chilled water conditions.
23 81 23 Part 2.2.5	CRAH-HUMIDITY-RANGE	C	Conforms. CRAH operating relative humidity range is 20% to 80% non-condensing, meeting spec.
23 81 23 Part 2.2.8	CRAH-REFRIGERANT-TYPE	NC	Non-Conformant. CRAH cooling coil utilizes R-407C refrigerant, but spec clause 2.2.8 explicitly mandates R-410A refrigerant for compliance with environmental policies.
23 81 23 Part 2.3.2.A	CRAH-FAN-POWER-DENSITY	C	Conforms. Fan power density is 0.09 W/(m3/h), which meets the specified limit of < 0.10 W/(m3/h).
23 81 23 Part 2.3.2.B	CRAH-MOTOR-UNIT	NC	Non-Conformant. CRAH fan motor power is listed as '7.5 HP' in the datasheet. The spec requires motor power to be defined in electrical kilowatts (kW) (conversion: 7.5 HP * 0.746 = 5.6 kW).
23 81 23 Part 2.3.5.A	CRAH-VALVE-CV	NC	Non-Conformant. The valve flow coefficient (Cv) of the 2-way modulating control valve is missing from the CRAH component table.
23 81 23 Part 2.3.5.C	CRAH-VALVE-TYPE	NC	Non-Conformant. CRAH is provided with a standard 2-way modulating control valve. Spec requires a pressure independent control valve (PICV) for chilled water flow regulation.

CERTIFICATE OF CONFORMANCE

Factory Quality Assurance Department

We hereby certify that the equipment listed below has been manufactured, tested, and inspected in accordance with all active factory testing specifications and international standards.

Equipment Model: CryoCore CR-250
Serial Numbers: CRY-2026-2301 to 08
Test Standards: IEC 62040-3, IEC 62271-200, ISO 8528-5 as applicable.
Quality Inspector: Dr. H. Bose (Director of QA)
Date of Inspection: January 8, 2026

Quality Inspector Signature	Plant Manager Signature